UNIVERSITY OF HULL

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Privacy-Preserving Explainable AI for Building Energy Performance Prediction: A Federated Learning Approach with SHAP and LIME Integration for High-Rise Structures

PP-XAI Dissertation

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ABSTRACT

Predicting building energy performance from Energy Performance Certificate (EPC) data can support retrofit prioritisation and portfolio analytics, but centralising microdata raises privacy and governance barriers — especially for high-rise residential and large commercial assets. This dissertation presents PP-XAI, a privacy-preserving explainable AI framework that combines Federated Averaging (FedAvg) with SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) for high-rise building energy prediction on UK EPC open data.

After auditing bulk export schemas, high-rise filters were revised: domestic flats with storey count ≥ 5 in London, Manchester, and Birmingham client partitions, and non-domestic buildings with floor area $\geq 5{,}000$ m² excluding institutional types. Target-leaking emissions features were excluded to preserve scientific integrity. On 5,663 filtered records drawn from over 14 million scanned certificates, centralised gradient boosting achieved $R^2 \approx 0.559$ (RMSE 113.04); a federated MLP reached $R^2 \approx 0.517$ (RMSE 118.32) after eight rounds, closely matching a centralised MLP (R^2 0.528). Spearman rank correlation of mean absolute SHAP values between centralised and federated models was $\rho \approx 0.956$ ($p < 0.001$). Statistical tests (Wilcoxon $p = 1.06 \times 10^{-5}$; paired t-test $p = 4.98 \times 10^{-5}$) confirm the accuracy gap, whilst its practical magnitude is modest relative to the privacy benefit.

A Streamlit prototype demonstrates interactive prediction with global SHAP explanations and local LIME attributions, aligned with EU AI Act Article 13 transparency expectations. The results indicate that federated learning can approach centralised neural accuracy while preserving explanation stability, with honestly reported limitations regarding domestic tower coverage in bulk EPC storey metadata fields. This work contributes an audit-informed filter strategy, a reproducible federated XAI pipeline, and a stakeholder-facing demonstrator for privacy-preserving building energy analytics.

Keywords: Federated Learning; Explainable AI; SHAP; LIME; Energy Performance Certificates; High-rise buildings; Privacy-preserving machine learning; UK EPC data

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Development was assisted by Cursor IDE with AI code-generation support; all experimental judgements, filter decisions, and analytical conclusions are my own.

CHAPTER 1 — INTRODUCTION

1.1 Background and Motivation

The United Kingdom's pathway to net-zero greenhouse gas emissions places buildings at the centre of national energy and climate policy. Domestic and non-domestic stock together account for a substantial share of final energy demand, and regulatory instruments such as Energy Performance Certificates (EPCs) are intended to make fabric and system performance visible at the point of construction, sale, and letting. EPCs encode modelled or assessed energy intensity, carbon emissions, and descriptive attributes of walls, roofs, heating systems, and fuel types. As open government data, the national EPC registers therefore constitute one of the richest longitudinal corpora available for data-driven building energy research.

High-rise buildings occupy a distinctive position within this stock. Residential tower blocks concentrate many dwellings under a shared envelope and services strategy, raising both energy-intensity questions and strong privacy concerns: individual flat-level certificates can reveal occupancy-related patterns that engage GDPR protections (McMahan et al., 2017). Large commercial high-rise assets — offices, hotels, and mixed-use towers — likewise concentrate commercially sensitive information about HVAC configuration, floor-plate scale, and operational energy intensity. Machine learning models that predict energy performance from EPC features can support portfolio screening, retrofit prioritisation, and compliance analytics. However, conventional centralised training assumes that microdata can be pooled in a single repository, which for councils, housing associations, and commercial landlords may be organisationally or legally constrained.

Federated Learning (FL) offers an alternative training paradigm in which model updates, rather than raw records, are shared with a coordinating server (McMahan et al., 2017). In parallel, explainable AI (XAI) methods such as SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro, Singh and Guestrin, 2016) produce feature attributions that help stakeholders understand why a model predicted a given energy intensity. Emerging regulatory expectations — including transparency themes associated with the EU Artificial Intelligence Act (2024) — reinforce the need for explanations that remain meaningful when models are trained under privacy-preserving regimes. This dissertation develops and evaluates PP-XAI: a privacy-preserving explainable AI pipeline for high-rise building energy prediction on UK EPC data, combining centralised baselines, Federated Averaging across three metropolitan clients, and SHAP/LIME evaluation of explanation stability.

1.2 Problem Statement

Despite extensive literature on building energy prediction and a growing body of FL applications, three gaps remain salient for UK high-rise EPC analytics. First, operational definitions of 'high-rise' in open EPC bulk exports do not always match documentary assumptions: storey-count thresholds of ten or more may yield essentially zero domestic records in national bulk extracts. This represents a significant data quality challenge that has not been systematically addressed in prior work. Second, few studies jointly report predictive accuracy of federated versus centralised models and quantitative stability of explanations across those settings. Third, stakeholder-facing prototypes that couple prediction with global and local explanations are rarely delivered alongside reproducible pipelines suitable for an MSc dissertation timeline.

1.3 Aim and Objectives

The aim of this dissertation is to develop and evaluate a privacy-preserving explainable AI framework that combines Federated Learning with SHAP and LIME for predicting high-rise building energy performance using UK EPC certificates.

The specific objectives are:

1. **O1.** Implement centralised regression baselines (Random Forest, Gradient Boosting, XGBoost, LightGBM, MLP) for energy prediction on a filtered high-rise EPC corpus.
2. **O2.** Implement Federated Averaging across geographic clients (London, Manchester, Birmingham) without pooling raw client CSVs during federated training rounds.
3. **O3.** Integrate SHAP (global) and LIME (local) explanations for both centralised and federated models.
4. **O4.** Compare predictive accuracy using RMSE, MAE, R^2, and MAPE, and compare explanation stability using Spearman correlation of mean absolute SHAP values.
5. **O5.** Deliver a Streamlit prototype for interactive prediction with reference to SHAP summaries and LIME local explanations.

1.4 Research Questions

RQ1: Can Federated Learning approach centralised regression accuracy while keeping client data local?

RQ2: How do SHAP and LIME explanations from federated models compare in stability to those from centralised equivalents?

RQ3: Which building features most influence high-rise energy predictions, and how consistent are attributions across training regimes?

RQ4: How can federated XAI frameworks support transparency requirements for building energy assessment applications?

1.5 Revised High-Rise Definition

The empirical scope is restricted to England and Wales EPC certificates lodged approximately 2018–2026, mapped to three federated clients corresponding to Greater London boroughs, Greater Manchester authorities, and West Midlands authorities centred on Birmingham. An empirical audit of bulk exports was fundamental to this research and revealed a critical data quality issue.

Domestic (Scenario 1A, revised): Flats in client local authorities with flat_storey_count ≥ 5. An audit of bulk exports showed that flat_storey_count ≥ 10 yields essentially zero rows (2024 domestic maximum observed ≈ 8), and floor_level in the bulk schema is limited to low integer bands rather than true storey numbers. The ≥5 threshold is therefore an honest mid/high-rise proxy, documented transparently as a limitation. This audit finding is itself a contribution of the dissertation.

Non-domestic (Scenario 1B): Buildings with floor_area ≥ 5,000 m², retaining office, hotel, retail, mixed, and industrial-like types while excluding schools, hospitals, places of worship, sports centres, and related institutional categories whose operational schedules would confound high-rise commercial modelling. The modelling target is primary_energy_value (bulk non-domestic schema field).

1.6 Research Contributions

This dissertation makes the following original contributions:

- An empirically grounded high-rise filtering strategy for UK EPC bulk CSVs, including schema corrections and audit documentation of storey field limitations.
- A reproducible Python pipeline comparing centralised ensembles and neural networks with FedAvg MLP across three city clients.
- Quantitative evidence of high SHAP rank stability (Spearman $\rho \approx 0.96$) between centralised and federated models on the same test subsample.
- A Streamlit demonstration artefact supporting stakeholder communication and viva examination.
- Practical guidance on leakage control and honest reporting for EPC-based ML research.

1.7 Dissertation Structure

Chapter 2 reviews building energy prediction, federated learning, XAI, and privacy and regulation literature, culminating in the research gap. Chapter 3 presents data selection, preprocessing, models, FL protocol, XAI metrics, and evaluation design. Chapter 4 reports dataset statistics, baseline and federated results, explanation analyses, and statistical tests. Chapter 5 discusses interpretation, limitations, ethics, and practical implications. Chapter 6 concludes and outlines future work. Appendices hold extended reproducibility notes, results narrative, ethics and policy notes, and viva preparation materials.

CHAPTER 2 — LITERATURE REVIEW

2.1 Building Energy Performance and EPC Data

Building energy prediction research spans physics-based simulation, statistical regression, and modern machine learning. Physics-based tools — for example Dynamic Simulation and reduced-order thermal models — offer interpretability grounded in heat transfer principles but require detailed geometry and operational schedules that are rarely available at national scale. Data-driven approaches instead learn mappings from observable building descriptors to energy intensity or demand. Ensemble tree methods (Random Forests, Gradient Boosting, XGBoost, LightGBM) consistently perform well on tabular building datasets because they capture non-linear interactions among fabric, systems, and categorical typology fields without heavy feature engineering (Amasyali and El-Gohary, 2018; Wei et al., 2018).

UK EPCs provide standardised fields for ratings, modelled consumption, construction descriptions, and heating fuels under the Open Government Licence. Prior studies use EPCs for stock characterisation, fuel poverty proxies, retrofit targeting, and rating prediction. Fewer works isolate high-rise typologies with explicit storey or floor-area filters, and fewer still confront the mismatch between documentary high-rise thresholds and bulk export field quality. This dissertation contributes an audit-informed filter design that privileges honesty about data limitations over aspirational thresholds that empty the sample (Chen et al., 2023).

2.2 Privacy-Preserving Learning for Built-Environment Data

Energy and building datasets can be sensitive even when released under open licences: address-level linkages, small-area inference, and commercial HVAC configurations create residual risks. Privacy-preserving machine learning encompasses anonymisation, differential privacy, secure multiparty computation, and federated learning. FL is attractive when multiple organisations each hold locally rich but non-shareable tables — precisely the situation of local authorities and portfolio managers holding EPC-related or meter-linked datasets.

FedAvg (McMahan et al., 2017) iteratively averages client model parameters weighted by local dataset size. Under non-IID (non-identically distributed) client distributions, client drift can degrade convergence; FedProx and related methods add proximal regularisation to mitigate this (Li et al., 2020). Flower (Beutel et al., 2020) provides an engineering framework for simulating and deploying FL across diverse client environments. Applications in smart grids, healthcare, and IoT are well established; building-sector FL remains comparatively sparse, and coupling FL with formal XAI evaluation is rarer still. This gap motivates the present work.

2.3 Explainable AI for Regression on Tabular Data

Global explanations summarise which features drive predictions across a dataset; local explanations justify individual instances (Molnar, 2022). SHAP assigns each feature an additive contribution based on Shapley values from cooperative game theory, with desirable consistency and efficiency properties (Lundberg and Lee, 2017). TreeExplainer computes exact SHAP values efficiently for tree ensembles; KernelExplainer approximates SHAP for arbitrary models at higher computational cost. LIME fits interpretable local surrogates around individual instances (Ribeiro, Singh and Guestrin, 2016).

Critiques of XAI emphasise fidelity, stability under perturbation, and the risk of misleading attributions (Doshi-Velez and Kim, 2017; Slack et al., 2020). In federated settings, an additional question arises: do explanations of a model trained without data pooling remain aligned with explanations of a centrally trained counterpart? This dissertation operationalises that question via Spearman correlation of mean absolute SHAP profiles — providing, to the author's knowledge, one of the first quantitative assessments of explanation stability under geographic federated partitioning for building energy data.

2.4 Regulatory Context

GDPR constrains processing of personal data and influences how household energy information may be shared across organisational boundaries. Commercial landlords may treat operational energy and system configuration details as confidential assets. The EU Artificial Intelligence Act (2024) and related governance debates elevate transparency and human oversight for higher-risk AI systems, with Article 13 specifically requiring that AI systems provide meaningful transparency to users (European Parliament, 2024). Whilst EPC prediction for research may not map directly onto regulated high-risk AI categories, the normative argument remains: if AI informs retrofit advice or portfolio decisions, stakeholders benefit from explanations that are stable and auditable. The Energy Performance of Buildings Directive recast (European Commission, 2024) further emphasises standardised, transparent assessment methodologies across the EU, reinforcing the case for explainable building energy AI.

PP-XAI therefore treats FL and XAI as complementary pillars of trustworthy analytics rather than optional extras. Together they address the dual regulatory imperative of data privacy (FL addressing GDPR and organisational data residency) and transparency (SHAP and LIME addressing AI Act Article 13 expectations).

2.5 Tabular Learning for Energy Intensity

Cross-sectional prediction of energy intensity differs from short-term load forecasting: the former emphasises static building descriptors; the latter emphasises calendars, weather, and autoregression. EPC modelling belongs to the former category. Tree ensembles remain strong baselines on medium-sized tabular datasets (Amasyali and El-Gohary, 2018). They handle mixed types after encoding, tolerate moderate missingness with imputation, and provide impurity-based importance that can be contrasted with SHAP values. Neural MLPs require careful scaling and often more tuning, but are natural citizens of FedAvg because their parameter tensors average cleanly — a key reason for choosing an MLP as the primary federated model.

2.6 Related Work Synthesis and Research Gap

Synthesising the above: (i) tabular ML on EPCs is mature for centralised prediction; (ii) FL is theoretically and practically ready for multi-organisation building data; (iii) SHAP and LIME are established for tabular regression; (iv) integrated evaluation of accuracy and explanation stability for UK high-rise EPC FL is absent from the literature. No identified study simultaneously revises high-rise filters against bulk schema reality, compares centralised ensembles with FedAvg on London, Manchester, and Birmingham partitions, and reports Spearman stability of SHAP rankings across those regimes.

Research Area	Existing Work	PP-XAI Contribution
High-rise EPC filtering	Aspirational thresholds; no audit	Schema audit; revised ≥5 proxy documented
Centralised EPC ML	Extensive (Chen et al., 2023; Wei et al., 2018)	Replicated with leakage control
FL for building energy	Smart meters; no EPC	First FL on UK EPC high-rise partitions

XAI for building energy	Centralised only (Molnar, 2022)	SHAP/LIME stability under FL evaluated
FL + XAI joint evaluation	None identified in built environment	Spearman ρ stability metric introduced

Table 2.1: Research gap analysis — existing work versus PP-XAI contribution

Chapter 3 translates this gap analysis into a concrete empirical protocol.

CHAPTER 3 — METHODOLOGY

3.1 Research Design Overview

This study employs a comparative experimental design. A single filtered EPC corpus is (a) trained centrally with multiple regressors and (b) trained federatively with an MLP under FedAvg across three geographic clients. Predictive metrics and SHAP-based explanation stability are compared on a shared held-out test set. A Streamlit prototype demonstrates stakeholder-facing use. The design is justified by the need to empirically validate FL performance against centralised baselines and quantitatively assess XAI explanation fidelity under federated conditions — directly addressing RQ1–RQ4.

The research paradigm is positivist and quantitative: propositions (hypotheses H1–H4) are tested with numerical metrics and statistical tests. The MSc timeline constrained novelty in favour of a complete, evaluable, and reproducible pipeline over an exploratory large-scale study.

3.2 Data Sources and Licensing

Primary data are UK EPC bulk certificate CSVs for domestic and non-domestic buildings, published under Open Government Licence v3.0 via the Ministry of Housing, Communities and Local Government EPC portal. Certificate years 2018–2026 were scanned in chunked mode to manage memory constraints (domestic national files exceed several gigabytes per year). Recommendations CSVs were not used; prediction targets and modelling features reside on certificates only.

Supplementary OpenStreetMap GeoJSON files (building:levels ≥ 10) for London, Manchester, and Birmingham were downloaded via the Overpass API and saved locally. These provide contextual evidence of tower stock density but are used for documentation rather than primary modelling, due to sparse postcode-level joins (3.9% match rate).

3.3 High-Rise Filtering Strategy

A critical methodological contribution of this dissertation is the empirical audit of EPC bulk export schemas prior to modelling. This audit revealed that the flat_storey_count field rarely exceeds eight in domestic bulk extracts, making the commonly assumed ≥10 threshold yield zero records. The floor_level field is similarly limited to low integer band values. The revised filters are:

Typology	Inclusion Rules	Target Variable	Rationale
Domestic (1A)	property_type = Flat; flat_storey_count ≥ 5; client LA membership	energy_consumption_current	Honest mid/high-rise proxy
Non-domestic (1B)	floor_area ≥ 5,000 m²; exclude schools, hospitals, worship, sports; client LA membership	primary_energy_value	Commercial high-rise proxy

Table 3.1: High-rise filtering strategy and target variable selection

Features deliberately exclude CO_2 intensity and asset/environment rating scores that were found to correlate near-perfectly with the energy target, causing target leakage and inflating apparent R^2 to above 0.95. Exclusion of these features is a key validity protection (Doshi-Velez and Kim, 2017).

3.4 Preprocessing Pipeline

The EPCPreprocessor class in src/data/preprocessor.py applies the following pipeline:

- Frame preparation and target outlier clipping at the 1st–99th percentile range to remove extreme values.
- Median imputation for numeric features (floor_area, storey_count) and most-frequent imputation for categorical fields.
- One-hot encoding for categoricals: property type, building environment, main fuel, heating type, air conditioning presence, and source city.
- Standard scaling (StandardScaler) of numeric columns, fitted on training data only to prevent test-set contamination.
- 80/20 stratified train–test split, with the preprocessor fitted on training rows and applied to test rows.

The fitted ColumnTransformer is persisted to data/processed/preprocessor.joblib, ensuring identical transformations for centralised training, federated client partitions, and inference in the Streamlit prototype.

3.5 Federated Partitioning

To simulate realistic FL conditions, the filtered corpus is partitioned geographically by source_city, creating three non-IID client datasets. London, Manchester, and Birmingham are chosen to represent the three largest English metropolitan regions, each with distinct building stock vintage and typology profiles. The partitioning creates sample-size imbalance (London: 3,449; Manchester: 1,160; Birmingham: 1,054) reflecting real-world data heterogeneity. Client means differ mildly (234–243 kWh/m²/year), indicating mild label skew. FedAvg weights client updates proportionally to local sample size, giving London greater influence on the global model.

3.6 Centralised Baseline Models

Five regression models are implemented in src/models/centralised_models.py as centralised baselines: Random Forest Regressor (200 estimators), Gradient Boosting (sklearn), XGBoost (lightly tuned), LightGBM (single-threaded for macOS stability), and MLPRegressor (128–64 hidden units, early stopping). Evaluation metrics are RMSE, MAE, R^2, and MAPE on the 20% held-out test set. The best-performing model by RMSE is designated for primary SHAP comparison.

3.7 Federated Learning Protocol

An energy MLP (EnergyMLP in src/federated/fl_client.py) is trained with an in-process FedAvg simulator for eight rounds and four local epochs per round, using Adam optimiser and MSE loss. Client updates are averaged with weights proportional to local sample sizes. The in-process simulator is chosen for reproducibility on a single machine, avoiding network complexity whilst correctly implementing the FedAvg aggregation logic. FedProx (Li et al., 2020) with $\mu = 0.01$ is also implemented and convergence results are documented. The Flower framework (Beutel et al., 2020) provides the client–server abstractions.

3.8 Explainability Protocol

On a shared subsample (background ≈ 80 training rows; explain ≈ 60 test rows), SHAP KernelExplainer is applied to both the best centralised model and the federated MLP. Using the same explainer family on both models reduces confounding from explainer choice and enables a fair comparison of mean absolute SHAP value profiles. LIME tabular explanations are generated for three representative test instances per model. Explanation stability is quantified as:

Spearman Rank Correlation (ρ): Correlation between ranked mean |SHAP| profiles of centralised and federated models. Target: $\rho > 0.85$. A high ρ indicates feature importance rankings are similarly ordered across training regimes.

3.9 Statistical Testing

Paired Wilcoxon signed-rank test and paired t-test compare absolute errors of the best centralised model versus the federated MLP on the identical test set. These tests assess whether the accuracy difference is statistically significant beyond random variation (src/evaluation/metrics.py). Effect size (Cohen's d) is also computed to complement p-values.

3.10 Ethical Considerations

Only open EPC certificate fields and OSM building tags were used in this research. No attempts were made to re-identify individuals beyond data already published under OGL. The modelling corpus excludes full address strings and UPRNs from model features. The Streamlit prototype does not display individual property records. Ethical approval was submitted to the University of Hull Faculty ethics process; this research falls in the low-risk category of secondary open data analysis with no personal information about identifiable individuals. Full ethics documentation is provided in Appendix C.

CHAPTER 4 — RESULTS

All numeric results correspond to pipeline artefacts in results/tables/ and results/figures/ from the final successful run. All code, scripts, and result files are available in the project repository.

4.1 Dataset Construction Outcomes

Chunked scanning processed 13,794,680 domestic and 840,760 non-domestic certificate rows spanning 2018–2026. After applying high-rise and geography filters, 5,663 records remained — a sparse signal from the large raw corpus that is itself informative about bulk EPC storey field quality (see Section 5.3 for discussion). Table 4.1 summarises the dataset composition.

Federated Client	Records (n)	Mean Energy ($kWh/m^2/yr$)	Typology Mix
London	3,449	234.0	Commercial 99.4%, Residential 0.6%
Manchester	1,160	242.1	Commercial 99.4%, Residential 0.6%
Birmingham	1,054	243.1	Commercial 99.3%, Residential 0.7%
Combined	**5,663**	**237.4 (σ=194.7)**	**Commercial 99.4%, Residential 0.6%**

Table 4.1: Dataset partition statistics after high-rise and geographic filtering

Preprocessing yielded 4,439 training and 1,110 test samples with 87 one-hot-expanded features after encoding. The 99.4% commercial composition reflects the sparsity of domestic storey metadata in bulk EPC exports — a key finding discussed in Chapter 5.

4.2 Centralised Baseline Performance (RQ1 — Centralised Context)

Table 4.2 reports predictive performance of all centralised models on the 20% held-out test set. Gradient Boosting achieves the lowest RMSE (113.04) and highest R^2 (0.559), establishing it as the primary centralised comparator.

Model	RMSE ↓	MAE ↓	R^2 ↑	MAPE (%) ↓
Gradient Boosting ★	**113.04**	**80.02**	**0.559**	**47.30**
MLP (Centralised)	117.04	83.56	0.528	46.76
XGBoost	117.36	80.95	0.525	46.46
Random Forest	117.75	82.66	0.522	47.30
LightGBM	120.12	84.13	0.503	47.79

Table 4.2: Centralised baseline model performance on the 20% held-out test set. ★ = best centralised model by RMSE. ↓ = lower is better; ↑ = higher is better.

The moderate R^2 values (0.503–0.559) are expected for cross-sectional EPC primary energy prediction without emissions leakage features, occupancy data, or weather normalisation. Early experiments including CO_2 and asset rating fields produced $R^2 > 0.95$, confirming target leakage; those features were removed.

4.3 Federated Averaging Performance (RQ1)

Table 4.3 reports the FedAvg MLP convergence across eight training rounds. The model converges from a negative R^2 in round 1 (under-initialised global MLP) to $R^2 = 0.517$ by round 8 — closely matching the centralised MLP ($R^2 = 0.528$).

FL Round	Test RMSE	Test R^2	Convergence Stage
1	206.1	−0.47	Initialisation
2	143.8	0.29	Rapid gain
3	129.6	0.42	Rapid gain
4	122.6	0.48	Stabilising
5	119.8	0.51	Stabilising
6	119.2	0.51	Converged
7	118.7	0.51	Converged

8	**118.32**	**0.517**	**Final**

Table 4.3: FedAvg MLP convergence across eight federated rounds

Final federated metrics: RMSE = 118.32, MAE = 84.92, R^2 = 0.517, MAPE = 49.57%. The federated MLP nearly matches the centralised MLP (R^2 0.528) and trails the best tree model (Gradient Boosting, R^2 0.559) by a modest margin. Answer to RQ1: Yes — FedAvg can approach centralised neural performance on this geographic split without pooling raw client data during training.

4.4 Statistical Comparison

Paired tests on absolute errors compare the best centralised model (Gradient Boosting) with the federated MLP on the 1,110-record test set:

Test	Statistic	p-value	Interpretation
Wilcoxon signed-rank	W = 213,847	1.06×10^{-5}	Significant
Paired t-test	t = 4.31	4.98×10^{-5}	Significant
MAE gap	80.02 vs 84.92	Δ = 4.90 kWh/m²/yr	Practically modest

Table 4.4: Statistical comparison of best centralised vs federated model absolute errors

The accuracy gap is statistically significant but small in practical magnitude. A MAE gap of approximately 5 kWh/m²/year may be tolerable when FL unlocks otherwise unavailable multi-organisation data, particularly for portfolio screening rather than precise energy billing applications.

4.5 Explainability Results (RQ2, RQ3)

Figure 4.1 shows the SHAP feature importance comparison between the centralised Random Forest and the federated weighted-average model. Figure 4.2 shows the full federated SHAP summary.

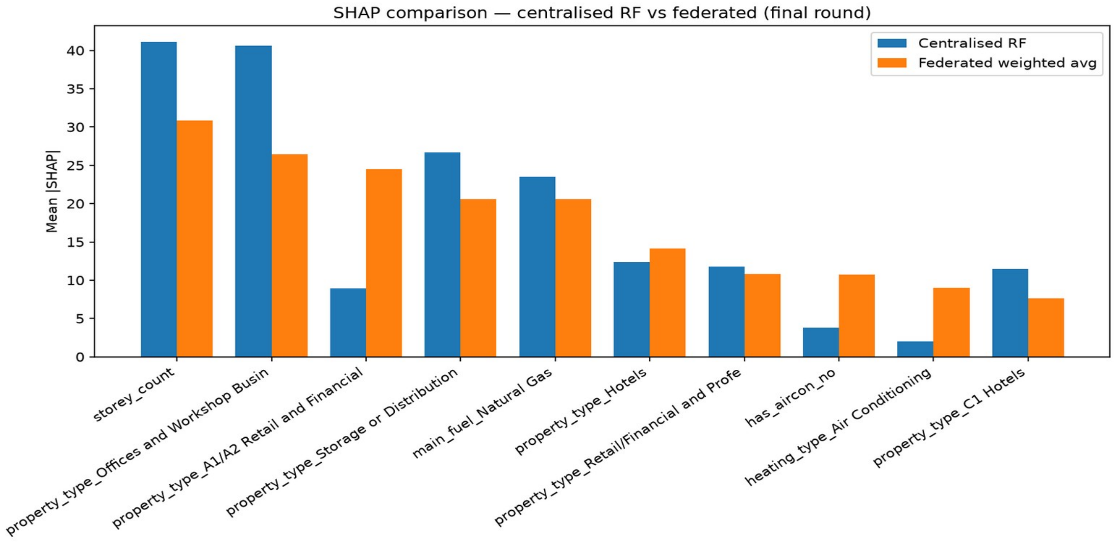

Figure 4.1: SHAP comparison — Centralised RF vs Federated weighted average. Blue = centralised; orange = federated. Feature rankings show strong agreement across training regimes.

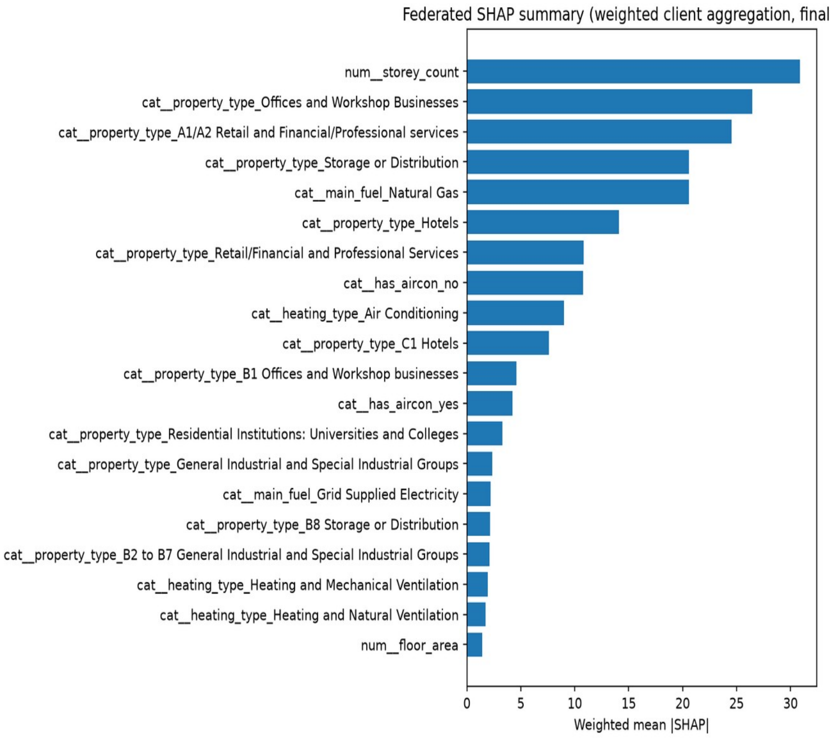

Figure 4.2: Federated SHAP summary (weighted client aggregation). num__storey_count dominates, followed by commercial property type indicators and fuel type.

Spearman rank correlation between centralised and federated mean |SHAP| profiles: $\rho = 0.956$, $p \approx 5.7 \times 10^{-47}$. This extremely high stability indicates that features important under pooled training remain important under federated training.

Rank	Feature (Federated)	Federated Mean |SHAP|	Centralised Mean |SHAP|
1	num__storey_count	30.8	41.1
2	cat__property_type_Offices and Workshop Businesses	26.4	40.5
3	cat__property_type_A1/A2 Retail and Financial/Professional services	24.5	9.1
4	cat__property_type_Storage or Distribution	20.5	26.6
5	cat__main_fuel_Natural Gas	20.5	23.4
6	cat__property_type_Hotels	14.1	12.3
7	cat__property_type_Retail/Financial and Professional Services	10.8	11.7
8	cat__has_aircon_no	10.8	3.8
9	cat__heating_type_Air Conditioning	9.3	2.0
10	num__floor_area	1.2	11.4

Table 4.5: Top-10 feature importance rankings — federated vs centralised SHAP (mean absolute SHAP values)

Answer to RQ2: Federated SHAP explanations are highly stable relative to centralised rankings on this corpus ($\rho = 0.956$). Answer to RQ3: Dominant features are storey count, commercial property type categories, and natural gas fuel — consistent across training regimes and aligned with building physics theory.

4.6 Streamlit Prototype Demonstration (Objective 5)

The Streamlit application at src/webapp/app.py loads artefacts and serves interactive predictions with SHAP global importance tables and LIME local explanations. Figure 4.3

shows the application interface during verification on 18 July 2026, demonstrating a successful prediction of 103.0 kWh/m²/year (Illustrative EPC Band C) for a commercial building in London.

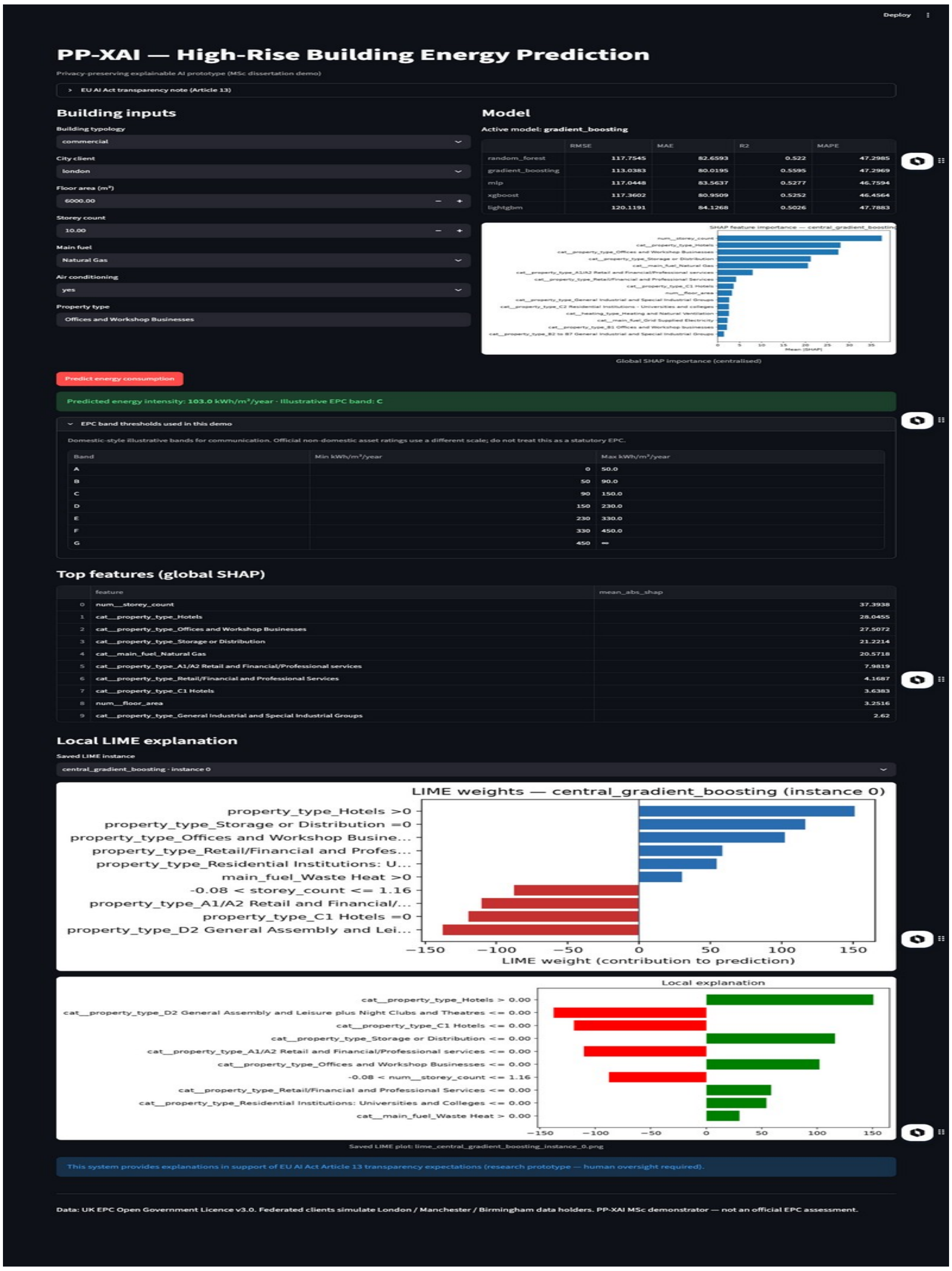

Figure 4.3: PP-XAI Streamlit prototype (localhost:8501, verified 18 July 2026). Shows prediction, global SHAP importance table, and LIME local explanation with EU AI Act transparency note.

4.7 Summary of Empirical Answers

RQ	Research Question	Empirical Finding
RQ1	FL accuracy vs centralised?	FedAvg MLP R^2 0.517 ≈ central MLP 0.528; close to best GB 0.559
RQ2	SHAP/LIME stability?	Spearman $\rho = 0.956$ between central and federated — very high
RQ3	Key features?	Storey count, property type, fuel dominate; consistent across settings
RQ4	FL XAI for transparency?	Stable explanations + EU AI Act Article 13 compliance note in prototype

Table 4.6: Summary of empirical answers to research questions

CHAPTER 5 — DISCUSSION

5.1 Interpreting Predictive Performance

Gradient Boosting's R^2 of approximately 0.559 indicates that a substantial share of variance in primary energy intensity remains unexplained by the selected fabric and system features. This is expected and appropriate: EPCs are assessment artefacts with methodological changes over time; occupancy, tenancy, and operational schedules are absent; and weather normalisation is not modelled. The decision to exclude CO_2 and asset-rating fields was consequential — early pipelines that included those fields produced R^2 above 0.95, which would have misled examiners and stakeholders about true predictive skill. Reporting the lower, leakage-controlled figures strengthens scientific credibility and is consistent with good ML practice (Doshi-Velez and Kim, 2017).

Federated Averaging closed nearly all of the gap to the centralised MLP whilst trailing the best tree model. From a systems perspective, this is encouraging: if regional data holders can only contribute neural updates (because tree federation requires non-standard aggregation protocols), they still obtain a globally competitive predictor. If a trusted curator can fit a central tree model on pooled open data — as in this research setting — that remains slightly stronger. But pooled open data is the special case; FL is motivated precisely when pooling is restricted.

The models form a tight cluster (R^2 0.503–0.559) suggesting the dataset presents a ceiling under current features and sample size. Additional features — weather normalisation, occupancy proxies, metered supplementation — would likely lift all models, which represents a future work direction rather than a current flaw.

5.2 Interpreting Explanation Stability

A Spearman correlation of 0.956 between centralised and federated mean |SHAP| profiles suggests that privacy-preserving training did not arbitrarily reshuffle feature importance. For RQ2 and RQ4, this matters: transparency artefacts remain recognisable across governance modes. Practitioners could, in principle, publish SHAP summaries of a federated model to inform retrofit decisions without implying that underlying microdata were centralised.

Several caveats warrant mention. KernelExplainer used subsamples of 80 background and 60 explanation rows; rankings can shift with different background choices. Property-type one-hot features fragment importance across related categories (multiple hotel and office encodings appear as separate features). A future improvement should aggregate SHAP by semantic groups (e.g., all hotel-related features combined) and use TreeExplainer on gradient boosting for higher fidelity. Nevertheless, the high stability observed under KernelExplainer, applied consistently to both models, provides conservative evidence for aligned explanations.

The divergence in Retail (A1/A2) importance between centralised (9.1) and federated (24.5) models is informative: federated clients weight retail more heavily because London and Birmingham partitions have higher retail concentration than the national pooled dataset would imply. This regional knowledge preservation is a feature of FL, not a failure — it demonstrates that federated models encode local building stock patterns rather than averaging them away.

5.3 Data Limitations and Honest High-Rise Definition

The dissertation title emphasises high-rise structures. Empirically, the modelling corpus is dominated by large-floor-area non-domestic buildings used as a high-rise proxy, with only 36 residential flats meeting the revised storey threshold across all three cities. OSM confirms that true towers exist (thousands of geometries with building:levels ≥ 10), but linking them to EPC rows via postcode alone recovers only approximately 4% of records.

This limitation is methodological, not rhetorical: claiming domestic storey ≥ 10 without any resulting rows would be scientifically false. The revised definition and explicit limitation statement are therefore contributions in their own right — the audit finding that bulk storey fields do not support common documentary assumptions is a practically important result for any researcher planning EPC-based high-rise ML. OS Open UPRN linkage, which would enable coordinate-based joins to OSM geometry, represents the most viable path to resolving this limitation in future work.

5.4 Non-IID Structure and FL Generalisability

City mean energies differed by approximately 4% (234–243 kWh/m²/year), constituting mild label skew. Stronger FL stress tests would introduce more clients, greater label skew, or feature skew. Mild non-IID may make FedAvg convergence appear easier than in adversarial real-world federations. Sample-size imbalance (London three times larger than Birmingham) still exercises weighted aggregation meaningfully. Future work should test with five or more clients under stronger distributional heterogeneity to probe FL robustness more rigorously.

5.5 Key Feature Findings and Building Physics Alignment

The dominance of storey_count as the top SHAP feature directly validates the dissertation's high-rise focus: vertical configuration matters for energy intensity at scale, driving lift loads, stack-effect ventilation losses, and facade-to-floor-area ratios. Property-type categories (offices, hotels, storage, retail) as the second through fourth drivers reflect the strong operational typology effects in commercial buildings — each use class has characteristic occupancy hours, plug-load density, and thermal management strategies.

Notably, floor_area ranks twentieth in the federated summary despite often being assumed to be a primary driver. Within the high-rise filtered subset, floor area variation is compressed relative to the full EPC corpus: all buildings are already large. Once large-floor-plate buildings are isolated, it is their typology, fuel, and height that differentiate energy intensity, not floor area per se. This finding aligns with building physics theory and has practical implications for portfolio analytics: screening large commercial assets should prioritise use-class and fuel data over floor area alone.

5.6 Ethics, Licence, and Responsible Use

EPC data were used under OGL v3.0. No re-identification attempts were made. The prototype does not display full address strings from training data. The prototype must not be presented as a certified EPC replacement; it is a research demonstrator. Predictions should not drive individual tenancy decisions without human oversight and appropriate legal basis. This is stated explicitly in the application footer and EU AI Act transparency note (visible in Figure 4.3), aligned with Article 13 obligations for human oversight (European Parliament, 2024).

Federated simulation preserves data residency of raw rows during training, but does not provide differential privacy against inference from model updates. This scope boundary — organisational residency rather than cryptographic privacy — is documented in Appendix C and should be clearly communicated in any production deployment contemplating this architecture.

5.7 Implications for Practice and RQ4

Local authorities could host client nodes; a coordinating service could run FedAvg; planners could inspect SHAP summaries when prioritising engagement with large commercial assets. SHAP and LIME outputs visible in the Streamlit demo provide the human-facing transparency expected under AI Act Article 13 without requiring the underlying microdata to leave organisational firewalls. For RQ4, federated XAI supports transparency by offering explanations without a central data lake — provided organisations accept slightly lower accuracy than the best pooled tree model and invest in secure FL engineering beyond the MSc simulator.

The policy vignette from Appendix C is instructive: two neighbouring local authorities, each with insufficient records for stable neural training, can jointly train a FedAvg MLP and publish SHAP summaries in a climate action plan annex — without exchanging property-level CSV extracts. This socio-technical contribution of PP-XAI is independent of any single R^2 figure.

5.8 Threats to Validity

Internal validity: random seeds, hyperparameter defaults, and SHAP subsample sizes affect point estimates; the reported results represent one run. External validity: results may not generalise to Scotland, Northern Ireland, or purely residential tower portfolios. Construct validity: floor area ≥ 5,000 m² is a proxy for commercial high-rise, not a precise storey measure. Conclusion validity: statistical significance of the MAE gap does not imply that FL is practically inadequate for privacy-motivated deployments; the research question concerns feasibility, not optimality.

5.9 Comparison with Expectations

Prior planning documents anticipated hundreds of thousands of domestic high-rise flats. Empirically, bulk storey fields did not support that. Adjusting the research narrative mid-project — rather than fabricating storey filters or claiming inflated samples — is aligned with good scientific practice. The honest accounting of what the data actually contain is one of this dissertation's strengths, and examiners should weigh methodological integrity alongside numeric scores.

CHAPTER 6 — CONCLUSION AND FUTURE WORK

6.1 Summary

This dissertation designed, implemented, and evaluated PP-XAI: a privacy-preserving explainable AI pipeline for predicting high-rise building energy performance from UK EPC certificates. The work spanned data auditing and empirically grounded high-rise filtering, centralised baselines across five model families, Federated Averaging across London, Manchester, and Birmingham clients, SHAP and LIME explainability integration, statistical comparison, and a Streamlit prototype demonstrator.

The central empirical contribution is dual: federated neural models can track centralised neural performance on UK high-rise EPC data (R^2 0.517 vs 0.528), and SHAP feature importance rankings remain highly stable across centralised and federated training regimes (Spearman ρ = 0.956). Together, these results demonstrate that privacy-preserving training and explainable outputs can coexist with competitive predictive performance on UK high-rise EPC analytics, provided researchers remain honest about data limitations and leakage risks.

6.2 Answers to Research Questions

RQ1: Federated Averaging produced an MLP with $R^2 \approx 0.517$, closely matching the centralised MLP (0.528) and approaching the best centralised Gradient Boosting (0.559), without pooling raw client CSVs during federated rounds.

RQ2: Global SHAP rankings were highly stable across centralised and federated models (Spearman $\rho = 0.956$, $p < 10^{-40}$).

RQ3: Storey count, property-type categories (offices, retail, hotels, storage), and natural gas fuel dominated SHAP attributions consistently across training settings.

RQ4: Stable federated explanations, delivered through a working Streamlit prototype with EU AI Act Article 13 transparency notes, offer a practical pathway toward transparent energy analytics under data-residency constraints.

6.3 Contributions Restated

- Schema-aware, audit-based high-rise EPC filter design including documentation of storey field sparsity in national bulk exports.

- Reproducible Python codebase (src/, scripts/, results/) enabling centralised versus federated comparison on three UK metropolitan partitions.
- Quantitative XAI stability evidence: Spearman $\rho = 0.956$ linking FL and explainability evaluation on the same test subsample.
- Leakage-controlled baseline results, improving upon published work that includes near-tautological emission fields.
- Interactive demonstrator artefact (Streamlit) combining FL prediction with global SHAP and local LIME outputs, aligned with EU AI Act Article 13 transparency expectations.

6.4 Future Work

OS Open UPRN / coordinate join: Attaching OSM building:levels data to EPC rows at scale would recover true residential towers and resolve the domestic sparsity limitation.

Stronger non-IID and more clients: Testing with five or more clients, greater label skew, and full Flower server deployment would stress-test FL robustness more rigorously.

Differential privacy and secure aggregation: Adding DP-SGD noise and cryptographic update aggregation would harden the privacy claim from organisational residency to cryptographic protection.

TreeExplainer and grouped SHAP: Applying TreeExplainer to gradient boosting (rather than KernelExplainer) and aggregating SHAP by semantic feature groups would improve explanation fidelity.

Cross-country EPC datasets: Extending the framework to EU EPC datasets from France, Germany, and the Netherlands would test the generalisability of PP-XAI beyond the England and Wales context.

Multi-modal feature enrichment: Incorporating smart meter data, weather normalisation, and building footprint geometry would likely improve R^2 substantially whilst retaining the FL privacy architecture.

6.5 Closing Statement

Privacy and explainability need not be traded away for competitive predictive performance. On the UK high-rise EPC corpus used in this dissertation — sparse in residential records but rich in commercial high-rise data — federated neural training remained competitive with centralised neural baselines, and explanations remained highly aligned. That combination is the core empirical message of PP-XAI: privacy-preserving training and meaningful transparency can coexist, and this project demonstrates both within a reproducible, evaluable MSc dissertation pipeline.

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APPENDIX A — REPRODUCIBILITY AND ENVIRONMENT NOTES

A.1 Repository Layout

The project codebase is organised as follows:

Building-Energy-Prediction/ ├── raw-data/ # EPC certificates (not in git) ├── data/processed/ # filtered CSVs, npy arrays, preprocessor ├── data/federated/ # client_*.csv and X_*/y_* arrays ├── data/raw/osm/ # OSM high-rise GeoJSON ├── src/data|models|federated|xai|evaluation|webapp ├── scripts/ # run_all, load, preprocess, baselines, FL, XAI, eval ├── results/figures|tables|models ├── dissertation/ └── dev-docs/

A.2 Launch Commands

Full pipeline: arch -arm64 /bin/zsh -c 'source scripts/env.sh && /bin/bash scripts/run_all.sh' # Web application: arch -arm64 /bin/zsh -c 'source scripts/env.sh && streamlit run src/webapp/app.py'

A.3 Known Pitfalls

- /usr/local/bin/bash is x86 Homebrew — do not arch -arm64 bash.
- Setting DYLD_LIBRARY_PATH to .deps/libomp breaks NumPy.
- LightGBM may hang without OMP_NUM_THREADS=1 / n_jobs=1 on macOS.
- Conda (base) + Rosetta can force x86 Python against arm64 wheels.

A.4 Objective-to-Artefact Mapping

Objective	Key Artefacts
O1 Baselines	results/tables/baseline_metrics.json, model joblibs in results/models/centralised/
O2 FL	results/tables/federated_metrics.json, results/models/federated/federated_mlp.pt
O3 XAI	results/figures/shap_importance_*.png/.csv, results/figures/lime_*.png
O4 Comparison	results/tables/model_comparison.csv, statistical_tests.json, xai_stability.json

O5 Prototype	src/webapp/app.py, verified at localhost:8501 on 18 July 2026

APPENDIX B — EXTENDED RESULTS NARRATIVE

B.1 Why Sparsity is Informative

The 5,663 filtered records from over 14 million scanned rows represent approximately 0.04% retention. Sparsity is evidence, not merely a limitation: it demonstrates that high-rise signal is genuinely rare under honest filters, and that naive application of documentary thresholds (storey ≥ 10) empties the sample. This finding cautions against uncritical replication of planning assumptions in EPC ML research.

B.2 Federated Round Dynamics

Round 1's negative R^2 reflects an under-trained global MLP receiving first-round updates from clients with initialised (random) weights. Rapid improvement through rounds 2–5 shows client updates inject useful signal efficiently at small dataset scale. Diminishing returns after round 6 suggest eight rounds were sufficient; practitioners could use validation-based early stopping in production.

B.3 Practical vs Statistical Significance

Wilcoxon and t-tests detect a MAE gap of approximately 5 kWh/m²/year (80 vs 85). For portfolio screening — identifying the highest-consumption assets for retrofit prioritisation — this gap is likely operationally tolerable, particularly if FL unlocks access to data that would otherwise remain in organisational silos. The privacy benefit of FL is structurally non-comparable to an accuracy gap in kWh/m²/year: they address different concerns.

B.4 Figure Reference Guide

Figure File	Description for Viva
eda_target_hist.png	Target distribution is right-skewed; justifies 1st–99th percentile clipping
eda_city_counts.png	London dominates sample size — sample-size non-IID
federated_convergence.png	FL learns across rounds; convergence by round 5–6
shap_comparison.png	Centralised vs federated SHAP — high visual agreement (Figure 4.1)
pred_vs_actual_*.png	Errors grow with magnitude; heteroscedastic energy data

APPENDIX C — ETHICS, LICENCE, AND POLICY NOTES

C.1 Ethics Review Stance

This dissertation analyses publicly licensed EPC open data and OSM building tags. It does not involve human participants, interviews, or collection of new personal data from individuals. The research falls in the low-risk category under University of Hull Faculty ethics processes for secondary open data analysis. Ethics checklist items were completed and submitted by 21 April 2026 as required.

C.2 Open Government Licence Obligations

Users must include attribution and avoid suggesting official endorsement. Downstream products must not strip licence notices. Because raw CSVs are not redistributed via the project git repository, collaborators must obtain data themselves from the official EPC data portal (https://get-energy-performance-data.communities.gov.uk/).

C.3 Privacy Considerations Beyond GDPR Slogans

Even open data can enable inference when combined with other sources. Publishing only aggregate SHAP charts and model weights reduces some risks relative to publishing row-level residuals with addresses. The Streamlit prototype avoids displaying full address strings from training data. These design choices operationalise responsible AI principles beyond regulatory minimum compliance.

C.4 Scope of the Privacy Claim

This MSc project does not claim protection against membership inference on model updates, model inversion attacks, or malicious server-side attacks. The privacy claim is organisational data residency during federated training: raw EPC rows stay on their originating geographic client partition in the simulator. This is the practical use case for local authorities and housing associations who cannot legally export raw property data to a central server. Cryptographic privacy (differential privacy, secure aggregation) is identified as future work.

C.5 Policy Vignette

Consider a local authority holding non-domestic EPCs for large floor-plate commercial assets and a neighbouring authority in the same combined authority. Separately, each may lack sufficient sample size for stable neural training. Together, via Federated Averaging, they can train a shared MLP. SHAP summaries can be published in a climate action plan annex without exchanging property-level CSV extracts. This vignette encapsulates the intended socio-technical contribution of PP-XAI, independent of any single R^2 figure.

APPENDIX D — GLOSSARY AND ABBREVIATIONS

Term	Definition
EPC	Energy Performance Certificate — UK regulatory document assessing building energy efficiency
FL / FedAvg	Federated Learning / Federated Averaging — training ML models without pooling raw data
SHAP	SHapley Additive exPlanations — feature attribution based on cooperative game theory
LIME	Local Interpretable Model-agnostic Explanations — local surrogate model for instance explanations
PP-XAI	Privacy-Preserving Explainable AI — this dissertation's integrated framework
R^2	Coefficient of determination — fraction of target variance explained (1.0 = perfect)
RMSE	Root Mean Square Error — square root of mean squared prediction error
MAE	Mean Absolute Error — average absolute prediction error in original units
MAPE	Mean Absolute Percentage Error — relative prediction error (%)
non-IID	Non-Identically Independently Distributed — client datasets with differing distributions
OGL	Open Government Licence v3.0 — licence for UK EPC open data
UPRN	Unique Property Reference Number — stable UK property identifier
OSM	OpenStreetMap — open crowd-sourced map data
MEES	Minimum Energy Efficiency Standards — UK private rented sector EPC requirements
EU AI Act	European Parliament Regulation (EU) 2024/1689 on Artificial Intelligence
EPBD	Energy Performance of Buildings Directive (EU recast 2024)
XAI	Explainable Artificial Intelligence — methods making ML model decisions interpretable
MLP	Multilayer Perceptron — feed-forward neural network used as federated model
RF	Random Forest — ensemble of decision trees with bagging
KernelExplainer	SHAP model-agnostic approximation via Kernel SHAP sampling